A Phase-Field Lattice Boltzmann Model for Ferrofluid Flow with Non-Collinear Magnetisation Dynamics

Rotational viscosity of a magnetic fluid in an alternating field

Abstract

A three-lattice multiple-relaxation-time lattice Boltzmann model is developed for two-phase ferrofluid flow in which the magnetisation is not collinear with the magnetic field. The conservative Allen–Cahn equation tracks the interface, the incompressible Navier–Stokes equations with Kelvin and rotational magnetic body forces govern the flow, and the magnetostatic field is obtained from a magnetic vector potential. The Martsenyuk–Raikher–Shliomis effective-field equation is integrated for the magnetisation, so that $\boldsymbol{M}\times\boldsymbol{H}\neq 0$ and the rotational viscosity effect is retained; this rules out the scalar-potential closure $\nabla\cdot(\mu\nabla\psi)=0$ used in existing phase-field ferrofluid models. It is further shown that a single-valued periodic scalar potential cannot carry a mean field along a periodic direction, which motivates the vector-potential formulation. All boundary closures are derived from discrete balances. The model is verified against the effective-field theory of Bacri *et al.* [Phys. Rev. Lett. **75**, 2128 (1995)] for plane Poiseuille flow of a magnetic fluid in an alternating field, over $0\le\xi_0\le 20$ and $0\le\omega\tau_B\le 2$, including the regime of negative rotational viscosity.

I. GOVERNING EQUATIONS

A. Interface and hydrodynamics

The interface is tracked by the conservative Allen–Cahn equation for the order parameter ϕ,

$$\frac{\partial\phi}{\partial t}+\nabla\cdot(\boldsymbol{u}\phi)=\nabla\cdot\left[M_\phi\left(\nabla\phi+\frac{4}{D}\phi(\phi-1)\hat{\boldsymbol{n}}_\phi\right)\right],\qquad \hat{\boldsymbol{n}}_\phi=\frac{\nabla\phi}{|\nabla\phi|},\tag{1}$$

with mobility M_ϕ and interface thickness D. Equation (1) is second order and mass conserving, and requires a single boundary condition. Density and viscosity follow the order parameter,

$$\rho=\rho_l+\phi(\rho_h-\rho_l),\qquad \eta=\eta_l+\phi(\eta_h-\eta_l).\tag{2}$$

The flow obeys the incompressible Navier–Stokes equations with three body forces,

$$\nabla\cdot\boldsymbol{u}=0,\qquad \rho\left[\frac{\partial\boldsymbol{u}}{\partial t}+\nabla\cdot(\boldsymbol{u}\boldsymbol{u})\right]=-\nabla p+\eta\nabla^2\boldsymbol{u}+\boldsymbol{F}_s+\boldsymbol{F}_m+\boldsymbol{F}_b,\tag{3}$$

where $\boldsymbol{F}_b$ is the imposed streamwise body force replacing the mean pressure gradient, and the surface tension force is

$$\boldsymbol{F}_s=\mu_\phi\nabla\phi,\qquad \mu_\phi=4\beta\phi(\phi-1)\left(\phi-\tfrac{1}{2}\right)-\kappa\nabla^2\phi,\qquad \beta=\frac{12\sigma}{D},\ \ \kappa=\frac{3D\sigma}{2}.\tag{4}$$

B. Magnetic body force

For a magnetisation that is not aligned with the field the magnetic stress is not symmetric, and the body force retains an antisymmetric part:

$$\boldsymbol{F}_m=\underbrace{\mu_0(\boldsymbol{M}\cdot\nabla)\boldsymbol{H}}_{\text{Kelvin}}+\underbrace{\frac{\mu_0}{2}\nabla\times(\boldsymbol{M}\times\boldsymbol{H})}_{\text{rotational}}.\tag{5}$$

In two dimensions, with $T=(\boldsymbol{M}\times\boldsymbol{H})_z=M_xH_y-M_yH_x$,

$$F_{m,x}=\mu_0(M_x\partial_xH_x+M_y\partial_yH_x)+\frac{\mu_0}{2}\partial_yT,\qquad F_{m,y}=\mu_0(M_x\partial_xH_y+M_y\partial_yH_y)-\frac{\mu_0}{2}\partial_xT.\tag{6}$$

The second term of (5) is the origin of the rotational viscosity: it vanishes identically when $\boldsymbol{M}\parallel\boldsymbol{H}$, and is the only route by which the magnetisation dynamics feeds back on the flow.

C. Magnetisation

The magnetisation is taken in the effective-field form of Martsenyuk, Raikher and Shliomis,

$$\boldsymbol{M} = \phi\,\varphi_p M_s L(\xi_e)\,\frac{\boldsymbol{\xi}_e}{\xi_e}, \qquad L(\xi) = \coth\xi - \frac{1}{\xi}, \tag{7}$$

in which the effective field $\boldsymbol{\xi}_e$ relaxes towards the true field $\boldsymbol{\xi} = \mu_0 \boldsymbol{m}\boldsymbol{H}/k_B T$ under the local vorticity $\boldsymbol{\Omega} = \frac{1}{2}\nabla\times\boldsymbol{u}$:

$$\frac{d\boldsymbol{\xi}_e}{dt} = \boldsymbol{\Omega}\times\boldsymbol{\xi}_e - \frac{1}{\tau_B}\frac{L_e}{L_e'}\left(1 - \frac{\boldsymbol{\xi}\cdot\hat{\boldsymbol{n}}_e}{\xi_e}\right)\hat{\boldsymbol{n}}_e + \frac{1}{2\tau_B}\left(\frac{\xi_e}{L_e} - 1\right)\left[\boldsymbol{\xi} - \hat{\boldsymbol{n}}_e(\boldsymbol{\xi}\cdot\hat{\boldsymbol{n}}_e)\right], \tag{8}$$

with $\hat{\boldsymbol{n}}_e = \boldsymbol{\xi}_e/\xi_e$, $L_e = L(\xi_e)$, $L_e' = dL/d\xi_e$. The second term relaxes the magnitude, the third the direction; the two limits $L_e/L_e' \to \xi_e$ and $\frac{1}{2}(\xi_e/L_e - 1) \to \xi_e/2$ recover Debye relaxation at weak field and the magnetic-torque rate $\mu_0 M_s H/6\eta = \xi/2\tau_B$ at strong field.

D. Magnetostatics

For a non-conducting fluid with no free currents,

$$\nabla\times\boldsymbol{H} = 0, \qquad \nabla\cdot\boldsymbol{B} = 0, \qquad \boldsymbol{B} = \mu_0(\boldsymbol{H} + \boldsymbol{M}). \tag{9}$$

Existing phase-field ferrofluid models close these with $\boldsymbol{M} = \chi\boldsymbol{H}$ and solve $\nabla\cdot(\mu\nabla\psi) = 0$. That closure is inadmissible here: it presupposes $\boldsymbol{M} \parallel \boldsymbol{H}$, whereas (7)–(8) give $\boldsymbol{M}\times\boldsymbol{H} \neq 0$, which is precisely the quantity that drives the effect under study.

Two potential formulations remain. The scalar potential $\boldsymbol{H} = -\nabla\psi$ gives $\nabla^2\psi = \nabla\cdot\boldsymbol{M}$, but is obstructed on a periodic domain: for any loop C winding once in a periodic direction $\hat{\boldsymbol{e}}$,

$$\oint_C \boldsymbol{H}\cdot d\boldsymbol{\ell} = -[\psi]_C = 0 \qquad\Longrightarrow\qquad \langle \boldsymbol{H}\cdot\hat{\boldsymbol{e}}\rangle = 0, \tag{10}$$

so a single-valued periodic ψ carries zero mean field along $\hat{\boldsymbol{e}}$ and cannot represent a field applied along the flow direction. The vector potential is not so obstructed. Writing $\boldsymbol{B} = \nabla\times(A\hat{\boldsymbol{z}})$ in two dimensions,

$$B_x = \frac{\partial A}{\partial y}, \qquad B_y = -\frac{\partial A}{\partial x}, \qquad \boldsymbol{H} = \frac{\boldsymbol{B}}{\mu_0} - \boldsymbol{M}, \tag{11}$$

which satisfies $\nabla\cdot\boldsymbol{B} = 0$ identically, and $\nabla\times\boldsymbol{H} = 0$ gives

$$\boxed{\nabla^2 A = -\mu_0\,(\nabla\times\boldsymbol{M})_z = -\mu_0\left(\frac{\partial M_y}{\partial x} - \frac{\partial M_x}{\partial y}\right)} \tag{12}$$

The source is the bound current $\nabla\times\boldsymbol{M}$ rather than the bound charge $\nabla\cdot\boldsymbol{M}$. For a uniform applied field $\boldsymbol{H}_0 \parallel \hat{\boldsymbol{x}}$ the corresponding potential is $A_0 = \mu_0 H_{0x} y$, which varies along the bounded wall-normal direction and is therefore representable on the lattice; this is the complement of (10). In the two-phase case $\nabla\times\boldsymbol{M} = \phi\,\nabla\times\boldsymbol{M}_f + \nabla\phi\times\boldsymbol{M}_f$, the second term being the interfacial bound surface current.

II. LATTICE BOLTZMANN FORMULATION

The D2Q9 velocity set is used for all three distributions, with $c = \Delta x/\Delta t$, $c_s^2 = c^2/3$,

$$\boldsymbol{e}_k = \begin{cases} (0,0), & k = 0, \\ c\left(\cos\frac{(k-1)\pi}{2},\ \sin\frac{(k-1)\pi}{2}\right), & k = 1,\dots,4, \\ \sqrt{2}\,c\left(\cos\frac{(2k-1)\pi}{4},\ \sin\frac{(2k-1)\pi}{4}\right), & k = 5,\dots,8, \end{cases} \qquad w_k = \tfrac{4}{9},\ \tfrac{1}{9},\ \tfrac{1}{36}. \tag{13}$$

Each lattice uses a multiple-relaxation-time collision, $\Lambda = \mathsf{M}^{-1}\mathsf{S}\mathsf{M}$ with the orthogonal moment matrix M of the reference model. Gradients of ϕ, $\boldsymbol{M}$ and $\boldsymbol{H}$ use the isotropic nine-point stencils

$$\partial_x\varphi = \frac{2(\varphi_{i+1,j+1} - \varphi_{i-1,j+1}) + 8(\varphi_{i+1,j} - \varphi_{i-1,j}) + 2(\varphi_{i+1,j-1} - \varphi_{i-1,j-1})}{24\,\Delta x}, \tag{14}$$

and the corresponding form for ∂_y. The normalisation $24\Delta x$ is required for consistency: the weights $(2,8,2)$ reduce to $(1,4,1)/12$, which returns unity for $\varphi = x$.

A. Phase field: the g lattice

$$g_k(\boldsymbol{x}+\boldsymbol{e}_k\Delta t, t+\Delta t) = g_k(\boldsymbol{x},t) - \left(\mathsf{M}^{-1}\mathsf{S}^g\mathsf{M}\right)_{k\ell}\left[g_\ell - g_\ell^{\mathrm{eq}}\right](\boldsymbol{x},t), \tag{15}$$

$$g_k^{\mathrm{eq}} = w_k\Lambda_\phi\phi + w_k\frac{\boldsymbol{e}_k\cdot\boldsymbol{u}}{c_s^2}\phi + w_k\frac{4M_\phi}{D}\phi(1-\phi)\frac{\boldsymbol{e}_k\cdot\hat{\boldsymbol{n}}_\phi}{c_s^2}, \qquad \phi = \sum_k g_k. \tag{16}$$

The three terms carry, in order, the order parameter, its advection, and the counter-term that keeps the interface profile at fixed thickness. The mobility sets the first-moment rates,

$$\frac{1}{s_3^g} = \frac{1}{s_5^g} = \frac{M_\phi}{c_s^2\Delta t} + \frac{1}{2}, \qquad \mathsf{S}^g = \mathrm{diag}\left(1.0,\, 1.1,\, 1.1,\, s_3^g,\, s_3^g,\, s_3^g,\, s_3^g,\, 1.2,\, 1.2\right). \tag{17}$$

B. Flow: the f lattice

$$f_k(\boldsymbol{x}+\boldsymbol{e}_k\Delta t, t+\Delta t) = f_k(\boldsymbol{x},t) - \left(\mathsf{M}^{-1}\mathsf{S}^f\mathsf{M}\right)_{k\ell}\left[f_\ell - f_\ell^{\mathrm{eq}}\right] + \left[\mathsf{M}^{-1}\left(\mathsf{I}-\tfrac{\mathsf{S}^f}{2}\right)\mathsf{M}\right]_{k\ell}F_\ell\Delta t, \tag{18}$$

$$f_k^{\mathrm{eq}} = w_k p + \rho c_s^2\left(\Gamma_k - w_k\right), \qquad \Gamma_k = w_k\left[1 + \frac{\boldsymbol{e}_k\cdot\boldsymbol{u}}{c_s^2} + \frac{(\boldsymbol{e}_k\cdot\boldsymbol{u})^2}{2c_s^4} - \frac{\boldsymbol{u}\cdot\boldsymbol{u}}{2c_s^2}\right], \tag{19}$$

$$F_k = \frac{\boldsymbol{e}_k-\boldsymbol{u}}{c_s^2}\cdot\left[c_s^2\nabla\rho\left(\Gamma_k - w_k\right) + \Gamma_k\left(\boldsymbol{F}_s + \boldsymbol{F}_m + \boldsymbol{F}_b\right)\right]. \tag{20}$$

The pressure-based equilibrium (19) and the density-gradient term in (20) are what allow large density ratios. Macroscopic quantities follow from the moments with the standard half-force correction,

$$\boldsymbol{u} = \frac{1}{\rho c_s^2}\sum_k \boldsymbol{e}_k f_k + \frac{\Delta t}{2\rho}\left(\boldsymbol{F}_s + \boldsymbol{F}_m + \boldsymbol{F}_b\right), \qquad p = \sum_k f_k + \frac{\Delta t}{2}c_s^2\,\boldsymbol{u}\cdot\nabla\rho. \tag{21}$$

Viscosity enters through a harmonic interpolation across the interface, which keeps the viscous flux continuous,

$$\frac{s_7^f}{1-0.5\,s_7^f} = \frac{\rho_l c_s^2\Delta t}{\eta_l} + \phi\left(\frac{\rho_h c_s^2\Delta t}{\eta_h} - \frac{\rho_l c_s^2\Delta t}{\eta_l}\right), \qquad s_8^f = s_7^f,\;\; s_4^f = s_6^f = \frac{8(2-s_7^f)}{8-s_7^f}. \tag{22}$$

C. Vector potential: the h lattice

Equation (12) is elliptic and is solved by embedding it as the steady state of a diffusion equation in a pseudo-time, with relaxation parameter ε:

$$\frac{\partial A}{\partial t} = \varepsilon\nabla^2 A + S, \qquad S = \varepsilon\,\mu_0\left(\nabla\times\boldsymbol{M}\right)_z. \tag{23}$$

Setting $\partial_t A = 0$ returns (12) for any ε, since the same ε multiplies both terms. The lattice equation is

$$h_k(\boldsymbol{x}+\boldsymbol{e}_k\Delta t, t+\Delta t) = h_k(\boldsymbol{x},t) - \left(\mathsf{M}^{-1}\mathsf{S}^h\mathsf{M}\right)_{k\ell}\left[h_\ell - h_\ell^{\mathrm{eq}}\right] + \Delta t\, w_k S, \qquad h_k^{\mathrm{eq}} = w_k A. \tag{24}$$

A Chapman–Enskog expansion of (24) recovers (23). Two contractions of the collision matrix are needed, both following from the structure of M (row 0 is unity, rows 3 and 5 are e_{kx}/c and e_{ky}/c):

$$\sum_k \Lambda_{k\ell} = s_0^h, \qquad \sum_k e_{k\alpha}\Lambda_{k\ell} = s_1^h e_{\ell\alpha}, \qquad s_1^h \equiv s_3^h = s_5^h. \tag{25}$$

That is, only the zeroth- and first-moment rates enter the recovered equation; the remaining rates act on ghost modes. The expansion gives

$$\frac{\partial A}{\partial t} = \underbrace{\Delta t\, c_s^2\left(\frac{1}{s_1^h} - \frac{1}{2}\right)}_{\varepsilon}\nabla^2 A + \underbrace{\left(\frac{s_0^h a}{\Delta t} + c_\Phi\right)}_{\mathscr{C}} S + \underbrace{\left[a\left(1-\frac{s_0^h}{2}\right) - \frac{\Delta t}{2}c_\Phi\right]}_{\mathscr{T}}\frac{\partial S}{\partial t}, \tag{26}$$

where c_Φ is the weight of the discrete source and a a possible shift in $A = \sum_k h_k + aS$. Matching to (23) requires

$$\varepsilon = \Delta t\, c_s^2 \left(\tau_h - \tfrac{1}{2}\right), \quad \tau_h = 1/s_1^h, \qquad \mathscr{C} = 1 \text{ (mandatory)}, \qquad \mathscr{T} = 0 \text{ (optional)}. \tag{27}$$

The pair $(c_\Phi, a) = (1,0)$ satisfies $\mathscr{C} = 1$ and is adopted; it leaves a residual $-\frac{\Delta t}{2}\partial_t S$ which is negligible whenever $\omega < \varepsilon k^2$, the same condition that makes the solve quasi-static. Any pair with $\mathscr{C} \neq 1$ rescales every computed field by that factor. Note that ε contains no permeability: (12) has constant coefficients, and ε is a pseudo-time relaxation rate of dimension $\mathrm{L}^2\mathrm{T}^{-1}$. The relaxation set is

$$\frac{1}{s_3^h} = \frac{1}{s_5^h} = \frac{\varepsilon}{c_s^2 \Delta t} + \frac{1}{2}, \qquad \mathsf{S}^h = \mathrm{diag}\Big(1.0,\, 1.1,\, 1.1,\, s_3^h,\, s_3^h,\, s_3^h,\, s_3^h,\, 1.2,\, 1.2\Big). \tag{28}$$

Since $\sum_k \boldsymbol{e}_k h_k^{\mathrm{eq}} = 0$, the first moment is purely non-equilibrium and returns the gradient of A directly, so that the induction follows without an auxiliary stencil:

$$A = \sum_k h_k, \qquad B_x = -\frac{1}{\tau_h c_s^2 \Delta t} \sum_k e_{ky} h_k, \qquad B_y = +\frac{1}{\tau_h c_s^2 \Delta t} \sum_k e_{kx} h_k. \tag{29}$$

III. BOUNDARY CONDITIONS

The domain is periodic in x and bounded by plane walls at $y = -\frac{1}{2}$ and $y = n_Y - \frac{1}{2}$, i.e. halfway between the boundary node and its image. Define the specular map σ, the bounce-back map $\bar{k}$, the incoming set and its weight,

$$\boldsymbol{e}_{\sigma(k)} = \boldsymbol{e}_k - 2(\boldsymbol{e}_k \cdot \hat{\boldsymbol{n}})\hat{\boldsymbol{n}}, \qquad \boldsymbol{e}_{\bar{k}} = -\boldsymbol{e}_k, \qquad K_{\mathrm{in}} = \{k : \boldsymbol{e}_k \cdot \hat{\boldsymbol{n}} < 0\}, \qquad W_{\mathrm{in}} = \sum_{k \in K_{\mathrm{in}}} w_k. \tag{30}$$

$$\text{For any lattice with } \boldsymbol{e}_k \cdot \hat{\boldsymbol{n}} \in \{-1, 0, 1\}: \qquad W_{\mathrm{in}} = \frac{c_s^2}{2} = \frac{1}{6} \text{ (D2Q9)}. \tag{31}$$

This follows from $\sum_k w_k (\boldsymbol{e}_k \cdot \hat{\boldsymbol{n}})^2 = c_s^2$, the summand being unity on K_{in} and on its mirror image and zero elsewhere.

A. Phase field

Mass conservation together with Young's law at the contact line requires

$$\hat{\boldsymbol{n}}_w \cdot \left[\nabla \phi_w - \frac{4}{D}\phi_w(1-\phi_w)\hat{\boldsymbol{n}}_\phi\right] = 0 \qquad \Longleftrightarrow \qquad \hat{\boldsymbol{n}}_w \cdot \nabla \phi_w = -\frac{4}{D}\cos\theta^{eq}\,\phi_w(1-\phi_w), \tag{32}$$

θ^{eq} being the equilibrium contact angle. Because (1) is second order, this single condition suffices. In the present single-phase verification $\phi \equiv 1$ and (32) is inactive; the g populations are streamed with specular reflection at the walls.

B. Flow

No-slip is imposed by halfway bounce-back,

$$f_k(\boldsymbol{x}_b, t + \Delta t) = f_{\bar{k}}^*(\boldsymbol{x}_b, t), \qquad k \in K_{\mathrm{in}}, \tag{33}$$

which places the zero-velocity plane at $y = -\frac{1}{2}$ and $y = n_Y - \frac{1}{2}$ and is consistent with the discrete cross-section factor used in Sec. VI.

C. Vector potential

Specular reflection alone transfers no potential across the wall,

$$\sum_{k\in K_{\rm in}} h_k(t+\Delta t) - \sum_{k\in K_{\rm out}} h_k^*(t) = 0, \tag{34}$$

because σ is a weight-preserving bijection $K_{\rm in} \leftrightarrow K_{\rm out}$; it therefore imposes $\partial A/\partial n = 0$, which is not the required condition. Augmenting it by a weighted increment,

$$h_k(\boldsymbol{x}_b, t+\Delta t) = h^*_{\sigma(k)}(\boldsymbol{x}_b, t) + B_w\, w_k, \qquad k \in K_{\rm in}, \tag{35}$$

transfers $B_w W_{\rm in}$ per step. Equating this to the Fickian flux $-\hat{\boldsymbol{n}}\cdot\boldsymbol{J} = \varepsilon\,\partial A/\partial n$ fixes the coefficient:

$$\boxed{B_w = \frac{\Delta t\,\varepsilon\,\gamma}{W_{\rm in}} = \frac{2\,\Delta t\,\varepsilon\,\gamma}{c_s^2}, \qquad \gamma \equiv \left.\frac{\partial A}{\partial n}\right|_{\rm wall}} \tag{36}$$

The wall datum follows from $\nabla A = \hat{\boldsymbol{z}}\times\boldsymbol{B}$: the normal derivative of A is the tangential induction, and continuity of the tangential field across the wall gives $\hat{\boldsymbol{t}}\cdot\boldsymbol{H} = \hat{\boldsymbol{t}}\cdot\boldsymbol{H}_0$. Hence

$$\gamma = \hat{\boldsymbol{t}}\cdot\boldsymbol{B} = \mu_0\left(\hat{\boldsymbol{t}}\cdot\boldsymbol{H}_0 + \hat{\boldsymbol{t}}\cdot\boldsymbol{M}\right), \qquad \hat{\boldsymbol{t}} = \hat{\boldsymbol{n}}\times\hat{\boldsymbol{z}}, \tag{37}$$

which for $\boldsymbol{H}_0 = H_{0x}(t)\hat{\boldsymbol{x}}$ and walls normal to $\hat{\boldsymbol{y}}$ reads

$$\gamma^{\rm bot} = -\mu_0(H_{0x} + M_x^w), \qquad \gamma^{\rm top} = +\mu_0(H_{0x} + M_x^w), \qquad M_x^w = \tfrac{3}{2}M_x\big|_1 - \tfrac{1}{2}M_x\big|_2, \tag{38}$$

the extrapolation placing γ at the wall rather than at the first node. *The applied field enters the calculation only here*; no additive field term appears anywhere in the interior, and $\boldsymbol{H}$ is obtained throughout from (11) and (29). Where A is prescribed instead of its gradient, the anti-bounce-back closure $h_k = -h_{\bar{k}}^* + 2w_k A_w$ applies; it is second-order accurate at a halfway wall.

D. Solvability

Summing the collision over k with (25) and $\mathscr{C} = 1$, the lattice mass changes by $\Delta t\, S$ per node and by $\Delta t\,\varepsilon\gamma$ per wall node. A steady state therefore exists if and only if

$$-\mu_0 \sum_{\boldsymbol{x}} (\nabla\times\boldsymbol{M})_z = \sum_{\rm walls} \gamma, \tag{39}$$

the discrete divergence theorem. With (38) this holds identically, the applied-field terms cancelling between the two walls. Residual mismatch between the nine-point stencil (14) and the wall extrapolation is removed by projecting the source onto the range of the discrete Laplacian,

$$\Delta_A = -\mu_0 \sum_{\boldsymbol{x}} (\nabla\times\boldsymbol{M})_z - \sum_{\rm walls}\gamma, \qquad (\nabla\times\boldsymbol{M})_z \;\leftarrow\; (\nabla\times\boldsymbol{M})_z - \frac{\Delta_A}{N}, \tag{40}$$

N being the number of nodes. In practice Δ_A/N remains at machine precision.

IV. COUPLING AND TIME INTEGRATION

Within one time step the fields are advanced in the order

$$\underbrace{A,\ \boldsymbol{B},\ \boldsymbol{H}}_{\text{from } h} \to \underbrace{\boldsymbol{M}}_{\text{MRSh}} \to \underbrace{S,\ \gamma}_{\text{from } \boldsymbol{M}} \to \underbrace{h \text{ sub-cycle}}_{N_{\rm sub}\text{ iterations}} \to \underbrace{\boldsymbol{F}_s, \boldsymbol{F}_m}_{\text{forces}} \to \underbrace{\boldsymbol{u},\ p}_{\text{moments}} \to \underbrace{f,\ g}_{\text{collide, stream}} . \tag{41}$$

The passes are separated because $\nabla\times\boldsymbol{M}$, the projection (40) and the force stencils are non-local.

A. Sub-cycling of the potential

A has no dynamics of its own: (12) is an instantaneous constraint, and the pseudo-time in (23) is a solver device. The h lattice is therefore iterated N_{sub} times per fluid step with S, γ and H_{0x} held fixed, which replaces the relaxation time by $\tau_{\mathrm{relax}}/N_{\mathrm{sub}}$ while nothing else in the step advances.

B. Magnetisation integration

Equation (8) is advanced with RK4 in N_{sub}^{M} sub-steps per lattice step. The field is *interpolated linearly across the step* rather than frozen,

$$\boldsymbol{H}(t_{n-1}+s) = \left(1-\frac{s}{\Delta t}\right)\boldsymbol{H}(t_{n-1}) + \frac{s}{\Delta t}\boldsymbol{H}(t_n), \qquad 0 \le s \le \Delta t. \tag{42}$$

Freezing $\boldsymbol{H}$ over the sub-steps is a first-order treatment of the explicit time dependence inside a fourth-order integrator; at $\omega\Delta t = 0.1$ it biases g by up to 34%, whereas (42) recovers the exact result to five decimal places.

V. PARAMETER SELECTION

The scheme carries three numerical time scales which must be correctly ordered against the physical ones. With $L = n_Y$, $\nu = \eta/\rho$ and ε chosen to minimise the potential relaxation time,

$$t_{\mathrm{visc}} = \frac{\rho L^2}{\pi^2 \eta} \qquad \text{momentum diffusion across the channel,} \tag{43}$$

$$\tau_{\mathrm{relax}} = \max\left(\frac{1}{\varepsilon k_{\mathrm{min}}^2}, \frac{\varepsilon}{c_s^2} + \frac{1}{2}\right) \;\ge\; \frac{L}{\pi c_s} \qquad \text{potential relaxation; minimised at } \varepsilon = \frac{c_s L}{\pi}, \tag{44}$$

$$\tau_B \ \text{(lattice steps)} \qquad \text{Brownian time; sets } \omega = \frac{\omega\tau_B}{\tau_B}. \tag{45}$$

The lower bound in (44) is a causality bound: streaming moves information one site per step, so a wall condition cannot reach mid-channel in fewer than $L/\pi c_s$ steps, whatever ε or the relaxation rates.

A. Field resolution

Solving (23) with an oscillating wall flux $\gamma_0 \cos\omega t$ on $[-L/2, L/2]$ gives $A = \mathrm{Re}[\hat{A}e^{i\omega t}]$, $\hat{A}'' = (i\omega/\varepsilon)\hat{A}$, and the odd solution $\hat{A} = C\sinh ky$ with $\hat{A}'(L/2) = \gamma_0$:

$$B_x(y) = \gamma_0 \frac{\cosh ky}{\cosh(kL/2)}, \qquad k = \sqrt{\frac{i\omega}{\varepsilon}}, \qquad \frac{kL}{2} = \frac{\Lambda(1+i)}{\sqrt{2}}, \qquad \Lambda = \frac{L}{2}\sqrt{\frac{\omega}{\varepsilon}}. \tag{46}$$

Expanding for small Λ,

$$\text{amplitude loss} \simeq \frac{\Lambda^4}{8}, \qquad \text{phase lag} \simeq \frac{\Lambda^2}{2}, \qquad \Lambda^2 = \frac{\pi^2}{4}\frac{\omega\tau_{\mathrm{relax}}}{N_{\mathrm{sub}}}. \tag{47}$$

Only the amplitude matters: a phase shift common to $\boldsymbol{M}$ and $\boldsymbol{H}$ cancels in the period-averaged torque, whereas an amplitude error shifts the effective ξ_0. Since the loss is fourth order in Λ, modest sub-cycling suffices. Requiring $E\,\Lambda^4/8 \le \mathrm{tol}$ with $E = |d\ln g/d\ln\xi_0| \lesssim 2$ gives

$$N_{\mathrm{sub}} \;\ge\; \frac{\pi^2}{4}\frac{(\omega\tau_B)\,\tau_{\mathrm{relax}}}{\tau_B} \Big/ \sqrt{\frac{8\,\mathrm{tol}}{E}}. \tag{48}$$

B. Flow resolution

The torque $\boldsymbol{M}\times\boldsymbol{H}$ is invariant under $(\boldsymbol{M},\boldsymbol{H}) \to (-\boldsymbol{M},-\boldsymbol{H})$, so under a field of frequency ω it oscillates at 2ω with no component at ω. The flow response to that component is attenuated by $(1+(2\omega t_{\mathrm{visc}})^2)^{-1/2}$, so the profile is steady to the extent that

$$2\omega t_{\mathrm{visc}} \gg 1, \qquad \text{and} \qquad \Omega\tau_B \ll 1, \qquad n_{\mathrm{steps}} \;\ge\; 10\,t_{\mathrm{visc}}, \tag{49}$$

the second being the condition under which the rotational viscosity is linear in the shear rate, and the third ensuring the start-up transient has decayed.

C. Magnetisation scale

Eliminating the particle volume between $\tau_B = 3\eta V/k_BT$ and $\xi_0 = \mu_0 m H_0/k_BT$ fixes the saturation magnetisation, and hence the ratio that controls how far the induced field is from negligible:

$$M_s = \frac{3\eta\,\varphi_p\xi_0}{\mu_0\tau_B H_0}, \qquad \frac{\varphi_p M_s}{H_0} = \frac{3\eta\,\varphi_p\xi_0}{\mu_0\tau_B H_0^2}. \tag{50}$$

Relation (50) is not optional: it is what makes g dimensionless and of order unity, and violating it rescales the measured rotational viscosity by the factor by which it is broken.

D. Values used

quantity	symbol	value	set by
grid	$n_X \times n_Y$	50×50	–
density, viscosity	ρ_h, η_h	1.975, 0.05	–
particle fraction	φ_p	0.2	material
Brownian time	τ_B	20 steps	numerical scale
field amplitude	H_{0x}	1.73	sets $\varphi_p M_s/H_0$ via (50)
potential relaxation	ε	9.2	$c_s n_Y/\pi = 9.19$
sub-cycles, potential	N_{sub}	64	(48)
sub-steps, magnetisation	N^M_{sub}	8	converged; see Sec. VI D
body force	F_b	2×10^{-7}	$\Omega\tau_B \simeq 10^{-3}$
steps	n_{steps}	200 000	$20\,t_{\mathrm{visc}}$
derived			
viscous time	t_{visc}	10 005 steps	(43)
potential relaxation	τ_{relax}	27.6 steps	(44)
cross-section factor	S	20 837.50	$\sum_j (j+\frac{1}{2})(n_Y - \frac{1}{2} - j)$

At $\omega\tau_B = 2$ these give $\omega\tau_{\mathrm{relax}}/N_{\mathrm{sub}} = 0.043$, $\Lambda^2 = 0.106$, an amplitude loss of 1.4×10^{-3}, and $2\omega t_{\mathrm{visc}} = 2001$.

VI. VERIFICATION

Plane Poiseuille flow of a magnetic fluid in an alternating field parallel to the flow is simulated and the rotational viscosity coefficient extracted. For a parabolic profile the discrete flow rate gives

$$\eta_{\mathrm{eff}} = \frac{\rho\, S F_b}{2Q}, \qquad Q = \sum_j \rho_j u_{x,j}, \qquad g = \frac{2}{3\varphi_p}\,\frac{\eta_{\mathrm{eff}} - \eta_0}{\eta_0}, \tag{51}$$

the last from $\Delta\eta = \frac{3}{2}\eta\,\varphi_p g$. Consistency of the force implementation with this definition follows from substituting $\mu_0\langle \boldsymbol{M}\times\boldsymbol{H}\rangle = -6\eta\,\varphi_p g\boldsymbol{\Omega}$ into (6): with $\Omega = -\frac{1}{2}u_x'$,

$$\frac{\mu_0}{2}\partial_y\langle T\rangle = \frac{3}{2}\eta\,\varphi_p g\,u_x'' \qquad\Longrightarrow\qquad 0 = \eta\left(1 + \tfrac{3}{2}\varphi_p g\right)u_x'' + F_b. \tag{52}$$

The reference is the effective-field theory, Eq. (8) integrated at fixed uniform vorticity over a whole number of field periods. The step size is capped by the stiffest rate, $L_e/L_e' \to \xi_e^2/\tau_B$, so that $h\xi_0^2$ stays below the RK4 stability limit of 2.785; this and the whole-period window are both required, and relaxing either shifts the reference curve by several per cent.

ξ_0	$\omega\tau_B$	η_{eff}	g_{LBM}	g_{theory}	ratio	swing	parabola resid.
0.0	0.0	0.05000010	+0.0000	+0.0000	–	6.2×10^{-6}	1.9×10^{-16}
1.0	0.0	0.05213361	+0.1422	+0.1426	0.997	3.7×10^{-6}	2.6×10^{-5}
2.5	0.0	0.05746252	+0.4975	+0.4989	0.997	1.0×10^{-6}	8.3×10^{-5}
5.0	0.0	0.06139929	+0.7599	+0.7621	0.997	3.6×10^{-7}	1.2×10^{-4}
10.0	0.0	0.06331423	+0.8876	+0.8901	0.997	2.0×10^{-7}	1.3×10^{-4}
15.0	0.0	0.06389462	+0.9263	+0.9289	0.997	1.9×10^{-7}	1.4×10^{-4}
20.0	0.0	0.06417281	+0.9448	+0.9475	0.997	2.1×10^{-7}	1.4×10^{-4}
1.0	2.0	0.04984887	−0.0101	−0.0099	1.016	1.9×10^{-5}	6.9×10^{-6}
2.5	2.0	0.04909305	−0.0605	−0.0596	1.015	1.2×10^{-4}	4.3×10^{-5}
5.0	2.0	0.04697903	−0.2014	−0.1986	1.014	4.3×10^{-4}	1.6×10^{-4}
10.0	2.0	0.04518618	−0.3209	−0.3180	1.009	1.2×10^{-3}	4.2×10^{-4}
15.0	2.0	0.04614604	−0.2569	−0.2590	0.992	1.8×10^{-3}	5.3×10^{-4}
20.0	2.0	0.04717356	−0.1884	−0.1973	0.955	2.1×10^{-3}	5.7×10^{-4}

The static branch reproduces the theory to 0.3% uniformly over a twenty-fold range of field amplitude. The alternating branch agrees to 1.6% up to $\xi_0 = 10$, including the change of sign of g near $\xi_0 \simeq 1.3$ and the minimum near $\xi_0 \simeq 9$, and to 4.5% at the highest amplitude. A sweep over $\omega\tau_B \in \{0, 0.1, 0.5, 1, 1.5, 2\}$ at the same settings reproduces the full family of curves, including the progressive suppression of g with frequency and the onset of negative rotational viscosity for $\omega\tau_B \gtrsim 1$.

Three diagnostics support the extraction. The zero-field run returns $\eta_0 = 0.05000010$ against a nominal 0.05, so the reference is converged to 2×10^{-6}. The parabola residual stays below 6×10^{-4}, so (51) is applicable. The residual 2ω ripple in η_{eff} ("swing") is at most 2×10^{-3}, consistent with $2\omega t_{\mathrm{visc}} = 2001$, confirming that the velocity profile is steady and that no time averaging of the output is required.

VII. LIMITATIONS AND OPEN ITEMS

1. **Residual at high field.** A systematic deviation remains at $\omega\tau_B = 2$, growing with ξ_0 to 4.5% at $\xi_0 = 20$, together with a uniform 0.3% offset on the static branch. The latter is multiplicative and independent of ξ_0, and therefore originates in the viscosity extraction rather than in the magnetisation model. The former has not been attributed.

2. **Mesh independence.** All results are at $n_Y = 50$. A refinement study requires ε, N_{sub}, F_b and n_{steps} to be rescaled together, since $\varepsilon \propto n_Y$, $\tau_{\mathrm{relax}} \propto n_Y$, $t_{\mathrm{visc}} \propto n_Y^2$ and $\Omega \propto F_b n_Y$; comparing at fixed n_{steps} compares different degrees of convergence rather than different meshes.

3. **Two-phase verification.** The magnetic module has been verified only for $\phi \equiv 1$. The interfacial bound current $\nabla\phi \times \boldsymbol{M}_f$ in (12) is implemented but untested.

4. **Three dimensions.** The vector potential reduces to a single scalar only in two dimensions. In three dimensions $\boldsymbol{A}$ has three components and requires a gauge condition, with the Coulomb gauge giving three copies of (12) coupled through their boundary data.